
UNIVERSITY OF CAPE TOWN
Final Examination
Investment Decision Theory and Application
ECO4102Z/ECO4122Z

Time: Three(3)hours

Maximum Marks: 100

INSTRUCTIONS TO CANDIDATES

1. You have 10 minutes at the start of the examination in which to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made on this examination paper. You then have three hours to complete the paper.
2. You must not start writing your answers in the booklet until instructed to do so by the invigilator.
3. Mark allocations are shown in brackets [].
4. Attempt all 8 questions, beginning your answer to each full question on a new page.
5. Show all calculations where this is appropriate.
6. Write legibly using blue or black ink.
7. Formulae Sheet attached at the end of this question paper

1. John and Lisa Jones, South African residents, are reviewing their financial plan. The Jones, both 53 years old, have one daughter, 18 years old. With their combined after-tax salaries totaling R1,800,000 a year, they are able to meet their living expenses and save R450,000 after taxes annually. They expect little change in either their incomes or expenses on an inflation-adjusted basis other than the addition of their daughter's college expenses. Their only long-term financial goal is to provide for themselves and for their daughter's education. The Jones both wish to retire in 10 years.

Their daughter, a talented musician, is now entering an exclusive five-year college program. This program requires a R900,000 contribution, payable now, to the college's endowment fund. Thereafter, her tuition and living expenses, to be paid entirely by the Jones, are estimated at R720,000 annually.

The Jones's personal investments total R20,800,000, and they plan to continue to manage the portfolio themselves. They prefer "conservative growth investments with minimal volatility." One-third of their portfolio is in the stock of Lisa's employer, a publicly traded technology company with a highly uncertain future. The shares have a very low-cost basis for tax purposes. The Jones, currently taxed at 30 percent on income and 20 percent on net realized capital gains, have accumulated losses from past unsuccessful investments that can be used to fully offset R1,800,000 of future realized gains.

In 10 years, John will receive a distribution from a family trust. His portion is now R21.6 million and is expected to grow prior to distribution. John receives no income from the trust and has no influence over, or responsibility for, its management. The Jones know that these funds will change their financial situation materially but have excluded the trust from their current financial planning.

- (a) Construct the objectives and constraints portion of an investment policy statement for the Jones, addressing each of the following:
 - i. Return objective
 - ii. Risk tolerance
 - iv. Liquidity requirements
 - vi. Unique circumstances

[8 marks]

Ten years have passed. The Jones, now both aged 63, will retire this year. The distribution from John's family trust will occur within the next two weeks. The Jones' current circumstances are summarized below:

Personal Circumstances and Assets

- Pension income will total R1,800,000 a year and will not increase with inflation.
- Annual expenses will total R3,240,000 initially and will increase with inflation.
- Inflation is expected to be 2 percent annually.
- Their personal investments now total R18,000,000 (excluding trust distribution).

- The Jones will rely on this R18,000,000 portfolio to support their lifestyle and do not wish to reduce their level of spending.
- The Jones have health problems and neither is expected to live more than 10 years. All health care expenses will be covered by employer-paid insurance.
- The Jones' daughter is now financially independent, and the Jones' sole investment objective is to meet their spending needs.
- The Jones are not concerned with growing or maintaining principal. The income deficit may be met with both investment income and by invading principal.

Trust Distribution Assets

- The trust distribution totals R36,000,000 and will occur within the next two weeks. No tax liability is created by the distribution.
- The Jones will maintain separate accounts for their personal assets and the trust distribution.
- They do not plan to withdraw income or principal.
- Tax liabilities produced by these assets will be paid from this portfolio.
- The Jones plan to donate these assets to an arts society when the surviving spouse dies. They have made a minimum pledge of R46.8 million toward construction of a new building.
- An after-tax annual return of 5.4 percent is required over five years to meet the minimum pledge.
- The Jones are concerned only that a minimum gift of R46.8 million is available. The Jones assume that at least one of them will live at least five years and that neither will live more than 10 years.

Alternative portfolios for the Jones' consideration appear in Table 1 below:

Table 1: Alternative Allocations

Asset Allocation	Portfolio			
	A	B	C	D
Domestic large-cap stocks	14%	30%	40%	30%
Domestic small-cap stocks	3	5	10	25
Foreign stocks	3	5	10	25
Intermediate-term fixed income	70	60	30	20
Cash equivalents	10	0	10	0
Total	100%	100%	100%	100%
Expected annual return*	4.2%	5.8%	7.5%	8.5%
Annual standard deviation	6.0%	8.0%	13.0%	18.0%

*Nominal after-tax returns

(b) For this part, answer either (i.) or (ii.), **do not answer both.**

- (i.) Select and justify with three reasons the most appropriate of the four portfolios from the table above as an asset allocation for the Jones' R18,000,000.
- (ii.) Select and justify with three reasons the most appropriate of the four portfolios from the table above as an asset allocation for the Jones' R36,000,000 in trust distribution assets.

[5 marks]

[Total: 13 marks]

2. (a) Explain clearly the difference between a float-adjusted market-capitalization-weighted index and a market-value-weighted index. [2 marks]
- (b) State any four(4) categories of shares that are included in the free float restrictions when calculating the FTSE/JSE Africa Index Series. [2 marks]

Consider the three stocks in the following table. P_t represents price, in Rand, at time t , and Q_t represents shares outstanding at time t . Stock C splits two for one in the last period.

	P_0	Q_0	P_1	Q_1	P_2	Q_2
A	90	100	95	100	95	100
B	50	200	45	200	45	200
C	100	200	110	200	55	400

- (c) Calculate the rate of return on a market-value-weighted index of the three stocks for the first period ($t = 0$ to $t = 1$). [2 marks]
- (d) Determine the divisor for the market-value-weighted index in period 2. [1 marks]
- (e) Calculate the rate of return for the second period ($t = 1$ to $t = 2$). [2 marks]

[Total: 9 marks]

3. A risk averse investor makes decisions using a quadratic utility function:

$$U(w) = w + dw^2.$$

- (a) Derive an upper bound for d for this investor. [2 marks]
- (b) Explain why the investor can only use this utility function to make decisions over a limited range of wealth, w . Your answer should include a statement of this range.[3 marks]

The investor states that the upper limit of wealth where she can use this utility function is $w = \text{R}10,000$.

- (c) Determine the value of d in the investor's utility function. [1 mark]

The investor wins a prize of R2,500 in a gameshow. She is then offered the opportunity to exchange this prize for a larger prize of R6,000 if she can answer

one more question correctly. However, she will receive no prize at all if she gets the question wrong. She estimates her chances of answering the question correctly to be 50%.

- (d) Determine whether the investor should take this opportunity to exchange. [3 marks]
[Total: 9 marks]

4. (a) Define an "efficient portfolio" in the context of mean-variance portfolio theory. [1 mark]

Suppose a market with three risky assets: X , Y , and Z with the following properties.

	Expected Return(%)	Standard Deviation(%)
Asset X	10	40
Asset Y	12	50
Asset Z	18	65

The risk-free rate is 4 percent, and the correlation between assets X and Y (ρ_{xy}) is 0.50, the correlation between assets X and Z (ρ_{xz}) is 0.71 and the correlation between assets Y and Z (ρ_{yz}) is 0.50.

- (b) Determine the expected return and standard deviation of a portfolio of X , Y and Z that maximizes the Sharpe ratio (i.e. optimal portfolio). Assume no restrictions to short selling. [10 marks]
- (c) Investors can hold any portfolio consisting of asset X , Y and Z and also the risk-free asset. In addition, suppose a mutual fund offers an expected return of 15% and a standard deviation of 49%. All investors have standard mean-variance preferences. Should any investor invest in the mutual fund? Explain clearly. [4 marks]

[Total: 15 marks]

5. (a) Multi-factor models of security returns have received increased attention. The arbitrage pricing theory (APT) probably has drawn the most attention and has been proposed as a replacement for the capital asset pricing model (CAPM).

Briefly explain the primary differences between the APT and the CAPM. [5 marks]

Suppose that three stocks (A, B, and C) and two common risk factors (1 and 2) have the following relationship:

$$E(R_A) = (1.1)\lambda_1 + (0.8)\lambda_2$$

$$E(R_B) = (0.7)\lambda_1 + (0.6)\lambda_2$$

$$E(R_C) = (0.3)\lambda_1 + (0.4)\lambda_2$$

- (b) If $\lambda_1 = 4\%$ and $\lambda_2 = 2\%$, what are the prices expected next year for each of the stocks? Assume that all three stocks currently sell for R30 and will not pay a dividend in the next year. [6 marks]

- (c) Suppose that you know that next year the prices for Stocks A, B, and C will actually be R31.50, R35.00, and R30.50. Create and demonstrate a riskless, arbitrage investment to take advantage of these mispriced securities. What is the profit from your investment? You may assume that you can use the proceeds from any necessary short sale. [5 marks]

[**Total: 16 marks**]

6. Louise and Greg Bolus live in the South Africa and currently rent an apartment in Cape Town's central business district area, in Sea Point. During an initial discussion of the Bolus' financial plans, Greg Bolus makes the following statements to the Bolus' financial adviser, Travis Bradfield:

- "I have used the ChatGPT extensively to research the outlook for the housing market over the next five years, and I believe now is the best time to buy a house."
- "I do not want to sell any bond in my portfolio for a lower price than I paid for the bond."
- "I will not sell any of my company shares because I know my company and I believe it has excellent prospects for the future."

For each statement (a)–(c) identify the behavioral bias most directly exhibited.

Explain how each behavioral bias is affecting Greg's investment decision making. [9 marks]

7. (a) An investment manager manages a South African equity fund and he has been asked to allocate the constituents of the FTSE/JSE Top 40 index into growth and value stocks. Describe how you would allocate the companies into growth and value. [3 marks]
- (b) The following information relates to the performance of two investment trusts and their equivalent benchmark index over a three year period. The annual risk-free rate of return over this period was 4% per annum.

	Trust A	Trust B	Index
Annual return (% p.a.)	9.0	8.0	7.0
Standard deviation (% p.a.)	13.5	9.5	6.5
Correlation coefficient with index	0.36	0.75	1.00

- (c) Calculate any four composite performance measures for each trust. [6 marks]
- (d) Comment on the results from part (a), stating any limitations that apply to them [4 marks]

[**Total: 13 marks**]

8. Two investment managers each managing part of a large pension scheme have produced the following figures (in millions of rand) for year 1:

	Total Fund	Equities	Bonds	Real Estate	Cash
A	100	25	50	20	5
B	100	35	40	15	10
Constituents of Index (%)					
Index		30%	50%	20%	0%
	Performance in year	Equities	Bonds	Real Estate	Cash
A		6.0%	3.0%	7.0%	0.5%
B		4.5%	4.0%	6.5%	0.5%
Index		5%	2%	6%	

- (a) Conduct an attribution analysis for the two funds in year 1. [6 marks]

The following attribution analysis has been produced for years 2 and 3.

Year 2	Fund	Index	Sector	Stock
	Performance	Performance	Selection	Selection
A	8.0%	7.0%	-0.5%	1.5%
B	6.0%	7.0%	0.0%	-1.0%
Year 3	Fund	Index	Sector	Stock
	Performance	Performance	Selection	Selection
A	1.5%	3.0%	0.5%	-2.0%
B	4.5%	3.0%	2.0%	-0.5%

Both managers are operating under the same mandate with the objective of outperforming their benchmark by 1% p.a. over a three year period.

- (b) Comment on the performance over the three year period. [3 marks]

Currently the benchmark is rebalanced every three years and the fund managers can deviate from the benchmark as they see fit. The trustees have now decided to rebalance the benchmark quarterly and to limit the freedom of the fund managers to deviate from the benchmark.

- (c) Discuss the merits and demerits of these proposed changes. [4 marks]
- (d) Propose, with reasons, any other factors that would need to be addressed when these changes are made. [3 marks]

[Total: 16 marks]